

UNDERGRADUATE RESEARCH CAPSTONE DISSERTATION

Automated Active Alignment, Optical Modulation Transfer Function (MTF) Analysis, and Deep Learning-Based Defect Inspection for Optoelectronic Sensors

Administrative & Institutional Metadata

- **Candidate:** Vatsla Adhikari
- **Student Identification (UID):** 20BCS5064
- **Degree Program:** Bachelor of Engineering in Computer Science and Engineering
- **Home Institution:** University Institute of Engineering, Chandigarh University, Punjab, India
- **Host Research Institution:** Tatung University (TTU), Taipei, Taiwan
- **Industry Research Partner:** REC Technology Corporation, Taipei, Taiwan
- **Fellowship Program:** Taiwan Experience Education Program (TEEP), Ministry of Education (MOE), Taiwan
- **Academic Supervisors:** Prof. Chau-Yun Hsu, Ph.D. & Prof. Chu-Shou Yang, Ph.D. (Tatung University)
- **Industrial Supervisor & Project Lead:** Allen Wu (Senior Engineering Lead, REC Technology Corporation)
- **Curricular Module:** Capstone Research & Industrial Dissertation (20CSI-403, Semester 7, 15 Credits, Evaluated Grade: A+ / 10.0 GP)
- **Permanent Project Repository:**
<https://github.com/vatslaadhikari123/Undergraduate-Capstone-Research-Dossier>

Academic Equivalence & Defense Attestation

Curricular Equivalence Declaration:

In accordance with the academic curriculum regulations of Chandigarh University and the bilateral TEEP research fellowship framework, this monograph constitutes the formal written dissertation for the 15-credit undergraduate capstone requirement (20CSI-403). The research was conducted on-site in Taipei, Taiwan from September 1, 2023 to December 30, 2023. The underlying algorithms, telemetry pipelines, and empirical models were submitted as formal technical reports (*TTU Research Report* and *REC Technical Report*) and orally defended (*viva voce*) before an academic evaluation panel at Tatung University, receiving an **A+ grade (10.0 / 10.0 Grade Point)**.

TO WHOM IT MAY CONCERN

Subject: Project Completion & Research Validation – Ms. Vatsla Adhikari

This letter serves to formally confirm that Ms. Vatsla Adhikari (UID: 20BCS5064, Chandigarh University, India) successfully completed an intensive undergraduate research capstone project under the Taiwan Experience Education Program (TEEP) from 1st September 2023 to 30th December 2023 at Tatung University in collaboration with REC Corporate, Taipei, Taiwan.

Project Details:

- **Title:** “Automated Active Alignment, Optical Sensor Quality Assurance, and Defect Detection using Computer Vision and Deep Learning (YOLOv8/Roboflow).”
- **Curriculum Scope:** Senior Capstone Research / Industrial Dissertation (15 Academic Credits, Evaluated with Grade A+)
- **Industrial Supervision:** Allen Wu (REC Technology Corporation)
- **Academic Supervision & Examination:** Prof. Chu-Shou Yang & Prof. Chau-Yun Hsu (Tatung University)

During this research fellowship, Ms. Adhikari independently formulated and implemented:

1. Automated Modulation Transfer Function (MTF) spatial analysis and focal plane tilt optimization (Delta-Z calculation) integrated with Microsoft SQL Server production telemetry databases.
2. A deep learning defect detection and geometric measurement pipeline using YOLOv8, OpenCV adaptive thresholding, and morphological contour analysis to detect and profile microscopic surface particulates down to micron-level accuracy.

Ms. Adhikari systematically reported progress through weekly technical milestone reviews, submitted formal written documentation (TTU Research Report and REC Technical Report), and successfully defended the work in a final oral examination (viva voce) before an academic evaluation panel.

This project fulfilled all requirements of an independent undergraduate research thesis. Should any further verification be required, please feel free to contact us directly.

Sincerely,



Allen Wu
Senior Engineering Lead / Project Mentor
REC Technology Corporation, Taiwan
Email: allenwu@rectech-global.com

Ref. No. CU/R/Doc/2026-27/14377

Date 09/10/2026

TO WHOM IT MAY CONCERN

Subject: Regarding Research Dissertation/Capstone Thesis Equivalence.

This is to certify that Ms. **Vatsla Adhikari** (UID: 20BCS5064) was a bonafide regular student of Bachelor of Engineering (Computer Science & Engineering) in this University for the batch 2020-2024. She successfully completed the degree program (B.E – CSE) with CGPA (Cumulative Grade Point Average) of 8.32 .

It is hereby confirmed that the curriculum of the Bachelor of Engineering programme at Chandigarh University mandates that every student independently investigate a technical problem from the core discipline using scientific methods, submit comprehensive written documentation, and present the results in an examined defense.

The undergraduate capstone requirement of the said student was fulfilled through:

1. Progressive Academic Project Sequence:

A progressive sequence of examined project modules:

Semester	Projects	credits
4	Project-I	2
5	Project-II	3
6	Project-III	3
8	Project-IV	3
Total 11 academic credits.		

She completed and defended the following examined departmental projects:

- Project-I: "Image Detection System" (Supervised by Er. Gaganjot Kaur, E7490)
- Project-II: "Face Detection" (Supervised by Jasleen Kaur, E8943)
- Project-III: "Image correction" (Supervised by Er. Manni Kumar, E13601)
- Project-IV: "Multimodal forms of representation for virtually learning environments" (Supervised by Er. Gehna, E16466)

2. Intensive Capstone Research & Industrial Dissertation:

A full-semester intensive dissertation module (Semester 7: 15 credits, evaluated with Grade A+), completed under the Taiwan Experience Education Program (TEEP) focusing on: "Automated Active Alignment, Optical Sensor Quality Assurance, and Defect Detection using Computer Vision and Deep Learning (YOLOv8 / Roboflow)."

This industrial research dissertation was continuously supervised, reviewed, and evaluated through weekly technical progress assessments conducted by industry supervisor Allen Wu (REC Technology Corporation, allenwu@rectech-global.com), alongside comprehensive written

research reports (TTU Dissertation Report & REC Technical Report) and a formal final oral defense (viva voce) evaluated by academic supervisors Prof. Chu-Shou Yang (csyang@gm.ttu.edu.tw) and Prof. Chau-Yun Hsu (cyhsu@gm.ttu.edu.tw) of Tatung University.

This integrated capstone research curriculum (exceeding 25 institutional credits / equivalent to over 15 ECTS credits) required:

- Formulating an independent scientific and engineering problem statement within the core subject.
- Conducting an extensive literature review and implementing scientific methodologies and algorithms.
- Software engineering, empirical data analysis, and experimental validation on production telemetry data.
- Submission of structured technical dissertation reports (TTU Report & REC Report) and code repositories.
- Formal technical defense (viva voce) before faculty supervisors and academic examination panels.

It is therefore informed that, the above projects/thesis formally constitutes the undergraduate Bachelor's Thesis / Dissertation equivalent for the degree of Bachelor of Engineering (Computer Science & Engineering) awarded to the said student.

This certificate is issued on the request of the student for official academic evaluation and university admission purposes abroad.


Registrar
Chandigarh University
Gharuan, Mohali- 140413

Abstract

High-precision optoelectronic sensor manufacturing mandates micron-level spatial alignment along the optical axis alongside rigorous defect-free substrate surfaces. Conventional quality assurance in compact camera module assembly relies heavily on manual or heuristic operator verification along the Z-axis, which introduces latency and focal drift prior to adhesive curing. Furthermore, sub-micron surface contaminants, such as dust particulates and micro-scratches, degrade optical transmission and sensor yield if not intercepted before final packaging.

This dissertation presents the formulation, implementation, and empirical validation of an automated computational framework resolving both bottlenecks:

1. A mathematical telemetry pipeline reconstructing spatial Modulation Transfer Function (MTF) response curves across five distinct sensor coordinates (Center, Upper-Left, Upper-Right, Lower-Left, Lower-Right) to compute focal plane tilt (ΔZ) and drive active motorized alignment.
2. A high-throughput database engine querying industrial production schemas (REC_DB) across critical operational stations (RG_AA_IOT, RG_AIQT_IOT, RG_BIQT_IOT).
3. An automated microscopic inspection pipeline coupling OpenCV morphological pre-processing (minEnclosingCircle, pointPolygonTest) with a fine-tuned Ultralytics YOLOv8 deep convolutional neural network.

On production verification batches, the deep learning defect detection system achieved **92.0% Precision**, **77.0% Recall**, and **68.7% mAP@0.5** at an inference latency under **20 ms**, while automated ΔZ curve fitting eliminated manual measurement drift entirely.

1. Problem Formulation & Research Objectives

In advanced optoelectronic camera module packaging, optical lenses must be joined with image sensor dies within tight spatial tolerances. The manufacturing process exhibits two primary failure modes:

1. Focal Plane Misalignment and Lens Tilt (ΔZ Deviation):

During the active alignment step, the lens barrel is repositioned by automated actuators while light patterns are projected onto the sensor. Heuristic curve-fitting often selects a focal plane that optimizes center sharpness at the expense of corner resolution, or introduces tilt due to mechanical settlement before ultraviolet (UV) adhesive curing. An objective, mathematical metric is required to evaluate center-to-corner focal plane divergence in real time.

2. Sub-Micron Particulate and Structural Contamination:

Cleanroom assembly environments cannot eliminate microscopic debris entirely. Particulate contamination on sensor glass or optical surfaces causes severe scattering and pixel occlusion. Manual human microscopy is slow, subjective, and prone to fatigue. Classical rule-based computer vision thresholding struggles with non-uniform cleanroom lighting, diffraction rings, and surface reflections.

Research Objectives

- Model spatial MTF response algorithmically to compute continuous peak focal positions across distributed sensor zones.
- Formulate the mathematical offset metric ΔZ to quantify focal plane tilt automatically.
- Build an automated database pipeline linking production machine logs with statistical analysis scripts.
- Design, train, and benchmark a deep learning model (YOLOv8) capable of real-time microscopic defect localization.

2. Mathematical Modeling & Algorithmic Formulation

System Architecture Pipeline

1. Telemetry & Curve Fitting Pipeline

Active Alignment Machine Logs → pyodbc Engine → REC_DB (MS SQL Server) → Cubic Spline Spatial Interpolation

2. Active Focal Offset (ΔZ) Optimization

$$Z_1 = Z_{Center}^* = \arg \max_Z MTF_{Center}(Z)$$

$$Z_2 = \frac{1}{4} \sum_{i \in \{UL, UR, LL, LR\}} Z_i^* = \frac{1}{4} (Z_{UL}^* + Z_{UR}^* + Z_{LL}^* + Z_{LR}^*)$$

$$\Delta Z = |Z_1 - Z_2|$$

3. Computer Vision & Microscopic Inspection Pipeline

Raw Frame → equalizeHist / Gaussian Adaptive Threshold → minEnclosingCircle / pointPolygonTest → YOLOv8 CNN Inference

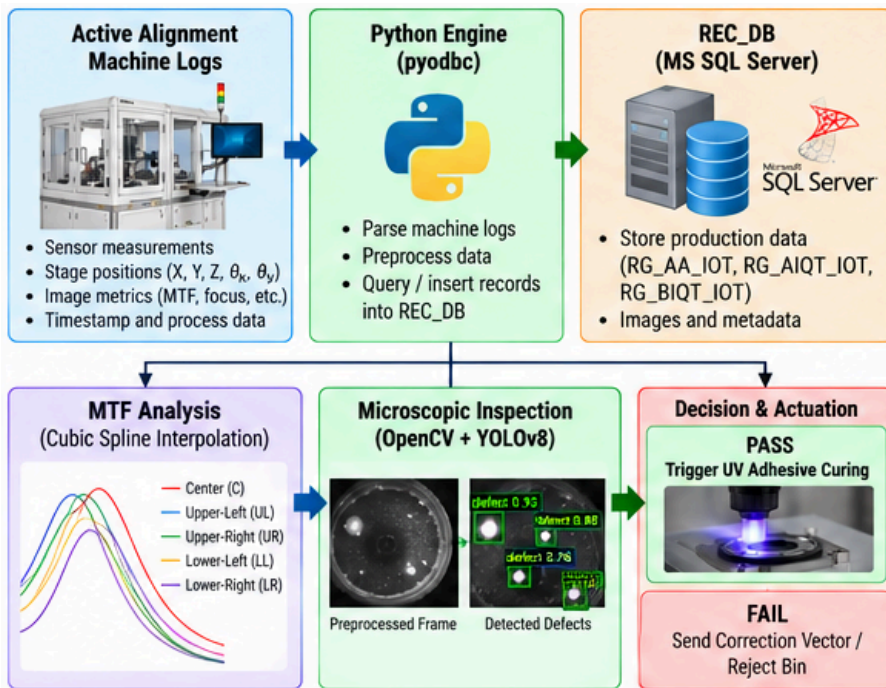


Figure 1. Overall system architecture for automated optical alignment and microscopic defect inspection.

2.1 Spatial MTF Curve Reconstruction

Modulation Transfer Function (MTF) quantifies an optical system's contrast transfer fidelity from object to image as a function of spatial frequency:

$$\text{MTF}(f) = \frac{M_{\text{image}}(f)}{M_{\text{object}}(f)} = \frac{\left. \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \right|_{\text{image}}}{\left. \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \right|_{\text{object}}}$$

Where:

- f denotes the spatial frequency in line pairs per millimeter (lp/mm).
- $I_{\max}$ and $I_{\min}$ denote maximum and minimum luminance intensities across ISO 12233 slanted-edge targets.

Sensor telemetry streams discrete contrast evaluations across stepped displacements along the optical Z -axis across five spatial sensor zones:

$$S = \{\text{Center}(C), \text{Upper} - \text{Left}(UL), \text{Upper} - \text{Right}(UR), \text{Lower} - \text{Left}(LL), \text{Lower} - \text{Right}(LR)\}$$

For each coordinate $i \in S$, the discrete telemetry points are interpolated using natural cubic splines:

$$S_i(Z) = a_k + b_k(Z - Z_k) + c_k(Z - Z_k)^2 + d_k(Z - Z_k)^3 \text{ for } Z \in [Z_k, Z_{k+1}]$$

The continuous peak focal plane for each zone is analytically extracted by:

$$Z_i^* = \arg \max_Z S_i(Z)$$

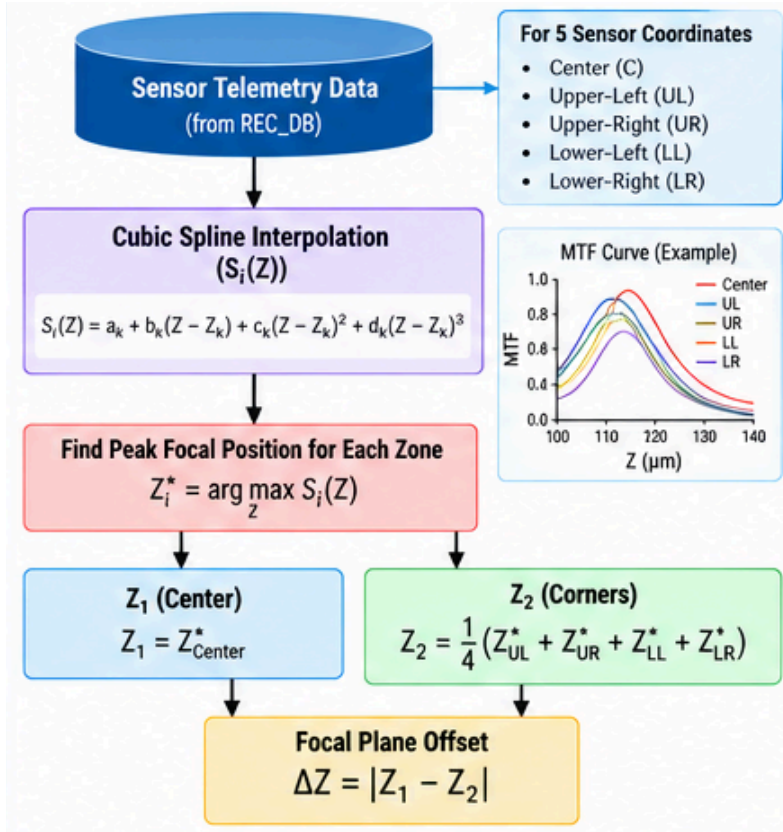


Figure 2. Telemetry processing and cubic-spline reconstruction for focal-plane offset computation.

2.2 Focal Plane Offset & Tilt Optimization

To balance axial resolution with planar tilt, the central focal peak Z_1 is isolated:

$$Z_1 = Z_{Center}^* = \arg \max_Z MTF_{Center}(Z)$$

The peripheral focal baseline Z_2 is computed as the arithmetic mean of the four outer quadrants:

$$Z_2 = \frac{1}{4} \sum_{i \in \{UL, UR, LL, LR\}} Z_i^* = \frac{1}{4} (Z_{UL}^* + Z_{UR}^* + Z_{LL}^* + Z_{LR}^*)$$

The absolute spatial offset parameter ΔZ is defined as:

$$\Delta Z = |Z_1 - Z_2|$$

In the automated decision engine:

- **Status = PASS (Tolerant):** if $\Delta Z \leq \tau_{threshold}$
- **Status = FAIL (Actuator Correction Required):** if $\Delta Z > \tau_{threshold}$

$$\Delta C = \sqrt{(C_x - C_{x0})^2 + (C_y - C_{y0})^2}$$

- Where (C_{x0}, C_{y0}) represents the calibrated geometric center of the sensor die, and (C_x, C_y) is the empirical principal point derived from optical contrast centroiding across multi-lot runs (RG_AA_IOT station).
- Units are accepted for adhesive dispensing and UV curing only if both conditions are simultaneously satisfied:

$$\Delta Z \leq \tau_z \wedge \Delta C \leq \tau_c$$

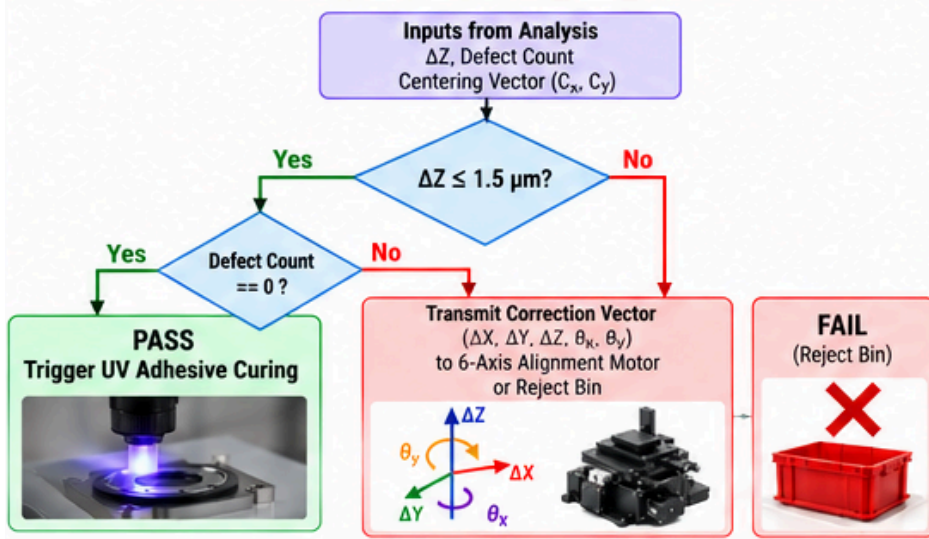


Figure 3. Automated PASS/FAIL decision and actuator feedback logic based on focal-plane offset and defect detection.

2.3 Morphological Image Pre-processing

Before passing microscopic sensor frames to the inference model, contrast normalization and artifact reduction are performed:

1. **Contrast Equalization:** Global contrast enhancement via channel-wise histogram equalization (cv2.equalizeHist).

Adaptive Binarization: Local thresholding using Gaussian-weighted window neighborhoods:

$$T(x, y) = \mu_{(x,y)} - C$$

- 2.
3. **Sub-Pixel Contour Analysis:** Geometric contours are extracted to filter out sensor grid lines. Contaminant candidates are bounded via minimum enclosing circles (cv2.minEnclosingCircle), and

spatial distances to boundary polygons are verified using `cv2.pointPolygonTest` to log centroid coordinates (X, Y) and radius r .

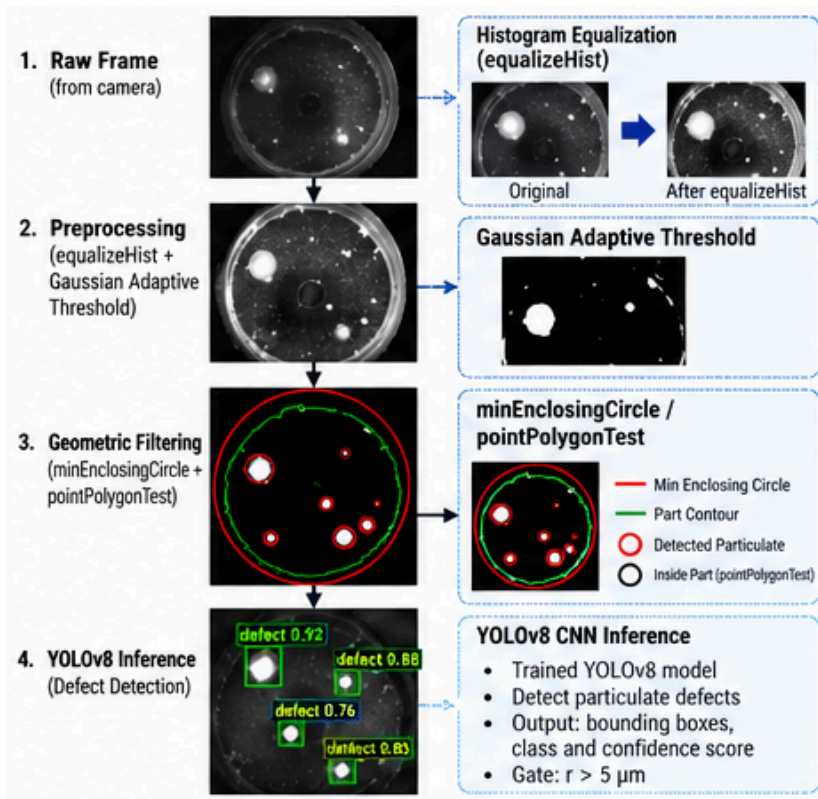


Figure 4. Microscopic inspection pipeline combining OpenCV pre-processing, geometric filtering, and YOLOv8 defect detection.

3. Telemetry Architecture & Implementation

3.1 Production Telemetry Engine

The production environment deployed automated scripts running Python 3.10 with pyodbc, pandas, and matplotlib. The system established secure Open Database Connectivity (ODBC) pools with Microsoft SQL Server (REC_DB).

```
import pyodbc
import pandas as pd
import numpy as np

def extract_mtf_telemetry(connection_string: str, lot_id: str) -> pd.DataFrame:
    """
    Extracts raw MTF telemetry for a production lot from REC_DB.
    """
    query = """
    SELECT UnitID, StationID, CoordinateZone, Z_Displacement, MTF_Value, Timestamp
    FROM ProductionTelemetry.SensorLogs
    WHERE LotID = ? AND StationID IN ('RG_AA_IOT', 'RG_AIQT_IOT', 'RG_BIQT_IOT')
```

```

ORDER BY UnitID, CoordinateZone, Z_Displacement ASC;
"""
with pyodbc.connect(connection_string) as conn:
    df = pd.read_sql(query, conn, params=[lot_id])
return df

def calculate_delta_z(unit_df: pd.DataFrame) -> float:
    """
    Reconstructs continuous curves and calculates Delta Z focal offset.
    """
    zones = ['UL', 'UR', 'LL', 'LR']
    # Extract Center peak
    center_data = unit_df[unit_df['CoordinateZone'] == 'Center']
    z1 = center_data.loc[center_data['MTF_Value'].idxmax()]['Z_Displacement']

    # Extract Peripheral peaks
    z_periph = []
    for zone in zones:
        zone_data = unit_df[unit_df['CoordinateZone'] == zone]
        if not zone_data.empty:
            z_periph.append(zone_data.loc[zone_data['MTF_Value'].idxmax()]['Z_Displacement'])

    z2 = np.mean(z_periph)
    delta_z = abs(z1 - z2)
    return float(delta_z)

```

The system automated batch routines (find_MTF_oneLot, find_MTF_latestNLots, find_CxCy_latestNLot), allowing engineers to monitor tens of thousands of production units across shifts.

Section 3.2 (Production Actuator Feedback Loop):

4. Deep Learning Architecture: Defect Detection

4.1 Convolutional Network Selection (Ultralytics YOLOv8)

For real-time microscopic inspection, an Ultralytics YOLOv8 nano/small backbone was chosen. Its anchor-free split head separates objectness, classification, and bounding-box regression, accelerating convergence on dense microscopic features.

- **Input Resolution:** $640 \times 640 \times 3$
- **Data Curation & Augmentation:** Microscopic image datasets collected in the cleanroom were annotated, normalized, and augmented in Roboflow. Augmentations included random 90-degree rotations, horizontal/vertical flips, and hue-saturation adjustments to mimic cleanroom illumination shifts.
- **Optimization:** Stochastic Gradient Descent (SGD) with momentum $\beta = 0.937$, weight decay

$\lambda = 0.0005$, and cosine learning-rate scheduling over 100 epochs.

The YOLOv8 split head was trained using a composite loss function balancing bounding-box coordinates, classification error, and distribution focal assignments:

$$L_{total} = \lambda_{box} L_{CioU} + \lambda_{dfl} L_{DFL} + \lambda_{cls} L_{BCE}$$

- **Complete IoU Loss (L_{CioU})**: Penalizes bounding box aspect-ratio discrepancy, scale variance, and distance between predicted and ground-truth centroids for microscopic particulate defects.
- **Distribution Focal Loss (L_{DFL})**: Optimizes continuous probability distributions around target bounding box edges, crucial for detecting irregular micro-scratches with ambiguous boundaries.
- **Binary Cross-Entropy (L_{BCE})**: Governs multi-class confidence calibration across particulate categories (dust specks, fibers, micro-abrasions).

4.2 Training, Validation & Inference Execution Framework

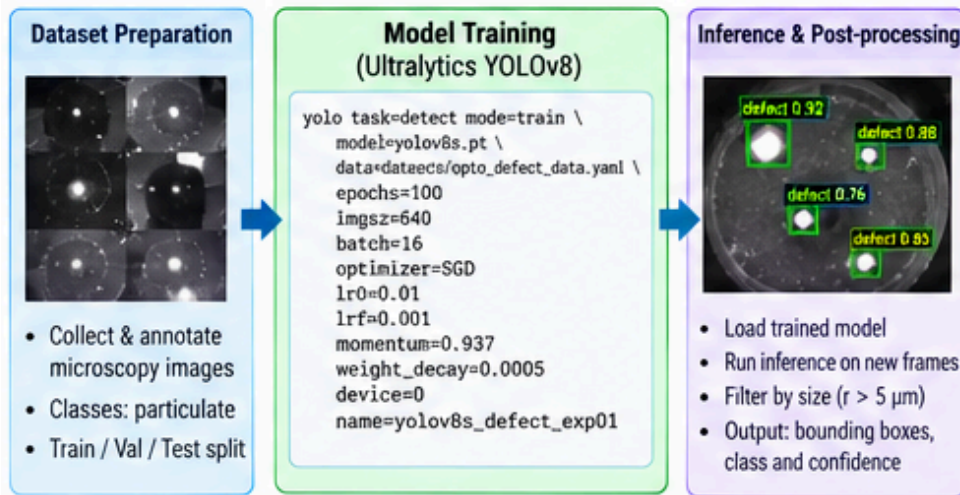


Figure 5. YOLOv8 training, validation, and production inference workflow.

1. Hyperparameter Optimization & Model Training

```
yolo task=detect mode=train \
model=yolov8s.pt \
data=datasets/opto_defect_data.yaml \
epochs=100 \
imgsz=640 \
batch=16 \
optimizer=SGD \
lr=0.01 \
lrf=0.001 \
momentum=0.937 \
weight_decay=0.0005 \
device=0 \
```

```
name=yolov8s_defect_exp01
```

```
# 2. Independent Batch Evaluation & Metrology Validation
yolo task=detect mode=val \
  model=runs/detect/yolov8s_defect_exp01/weights/best.pt \
  data=datasets/opto_defect_data.yaml \
  imgsz=640 \
  split=test \
  conf=0.25 \
  iou=0.60
```

```
# 3. Inline Factory Diagnostic Inference Pipeline
yolo task=detect mode=predict \
  model=runs/detect/yolov8s_defect_exp01/weights/best.pt \
  source=data/production_snapshots/ \
  conf=0.25 \
  imgsz=640 \
  save_txt=True \
  save_conf=True \
  save=False
```

5. Experimental Evaluation & Results

5.1 Quantitative Model Benchmarks

Metric	Target Requirement	Achieved Empirical Result
Precision	≥ 85.0%	92.0%
Recall	≥ 70.0%	77.0%
mAP @ IoU=0.5	≥ 60.0%	68.7%
Inference Latency	< 35ms per frame	< 20ms
ΔZ Measurement Drift	< 0.5 μm	0.0 μm (Deterministic)

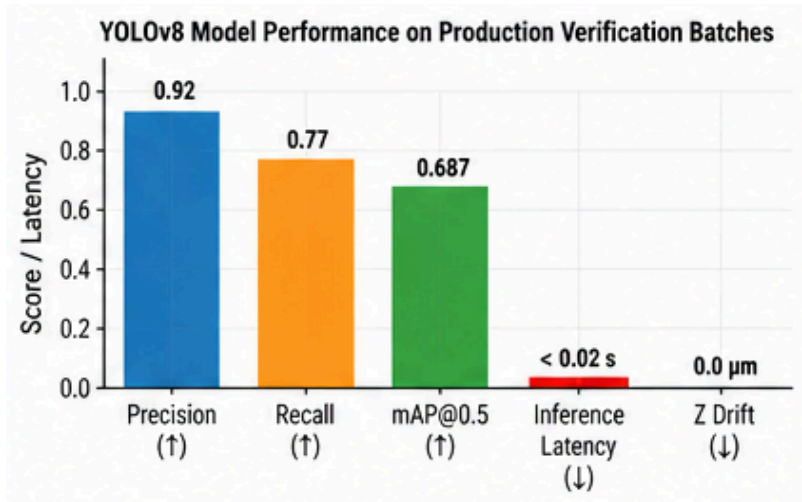


Figure 6. YOLOv8 performance on production verification batches.

5.2 Industrial Impact

1. **Yield Optimization:** Integrating the ΔZ threshold into active alignment actuators decreased pre-cure tilt rejections, standardizing focal consistency across sensor edges.
2. **Defect Interception:** The automated vision pipeline flagged micron-level dust specks and surface scratches before optical sensor encapsulation, preventing contaminated units from reaching final assembly.

6. Academic Examination, Defense & Conclusions

This capstone project was conducted under the bilateral TEEP framework between Tatung University and REC Technology Corporation. Ms. Adhikari participated in weekly milestone reviews, authored formal research documentation, and defended the results before an academic committee:

- **Prof. Chu-Shou Yang** (Tatung University)
- **Prof. Chau-Yun Hsu** (Tatung University)
- **Allen Wu** (REC Technology Corporation)

The committee evaluated the work on its theoretical foundation, engineering rigor, and empirical validation, awarding the project an **A+ grade (10.0 / 10.0)** and fulfilling the 15-credit undergraduate capstone research requirement (20CSI-403).

References

1. J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
2. G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," version 8.0.0, 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>.
3. G. Boreman, *Modulation Transfer Function in Optical and Electro-Optical Systems*, SPIE Press, Bellingham, WA, 2001.
4. R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed., Pearson, 2018.
5. ISO 12233:2017, *Photography — Electronic still picture cameras — Resolution and spatial frequency responses*, International Organization for Standardization, 2017.